

W3Link

Product Roadmap & Technical Assessment

Missing Features, Future Development Plan, and Security Review

May 2026

1. Vision Statement

W3Link is positioned as a gateway for humans to interact with the AI agents they have hired. It is a window very similar to the messengers we use every day, but designed for a different purpose: to help people stay in touch with both their human and non-human employees, facilitating interaction and collaboration.

W3Link serves three roles simultaneously:

1. **Corporate Messenger** -- private, secure enterprise communication
2. **AI-Based Project Control Tool** -- managing tasks, projects, and workflows through AI
3. **Window to the AI Agent Marketplace** -- discovering, hiring, and managing AI agents

2. Current Feature Status

The following table shows the current implementation status across web and mobile platforms, along with development priority for unfinished features.

Category	Feature	Web	Mobile	Priority
Messaging	Basic messaging	Done	Done	-
Messaging	Voice/Video calls	Done	Partial	High
Messaging	End-to-end encryption	Not Started	Not Started	Critical
Messaging	Threaded replies	Not Started	Not Started	High
Messaging	Message editing history	Not Started	Not Started	Medium
Messaging	Rich formatting (Markdown)	Not Started	Not Started	Medium
Messaging	Global message search	Not Started	Not Started	High
Messaging	Channels (public/private)	Not Started	Not Started	High
AI	Linda AI Secretary	Done	Done	-
AI	Agent Marketplace	Done	Done	-
AI	TransGuy Translation	Done	Done	-
AI	Linda Web Search	Done	Not Started	Medium
AI	Meeting Summarizer Agent	Not Started	Not Started	High
AI	Code Review Agent	Not Started	Not Started	Medium
AI	Onboarding Agent	Not Started	Not Started	Medium
AI	Agent-to-Agent Orchestration	Not Started	Not Started	Low

Management	Task Management	Done	Partial	Medium
Management	Project Boards	Done	Partial	Medium
Security	Audit Logs	Not Started	N/A	Critical
Security	Granular Permissions	Not Started	N/A	High
Security	Read Receipt Privacy	Not Started	Not Started	Medium
Platform	Notification Controls	Not Started	Not Started	High
Platform	Organization Wizard	Not Started	Not Started	Medium
Platform	Custom Domain Support	Not Started	Not Started	Low

3. Messenger Features Roadmap

3.1 End-to-End Encryption (E2EE) -- CRITICAL

End-to-end encryption is the highest priority security feature. All message content must be encrypted client-side before transmission.

- Implement Signal Protocol or similar proven cryptographic protocol
- Key exchange mechanism for establishing secure sessions
- Encrypted media storage for files, images, and voice messages
- Key rotation and device management for multi-device support
- Consider: libsignal-protocol-javascript as reference implementation

3.2 Threaded Replies

- Reply chains within conversations for organized discussions
- Thread notifications separate from main chat notifications
- Thread summary displayed in main chat view

3.3 Channels (Public & Private)

- Slack-style channels with topics and descriptions
- Channel permissions and roles (admin, member, read-only)
- Channel search and discovery within organization
- Announcement-only channels for company-wide broadcasts

3.4 Global Message Search

- Full-text search across all conversations and channels
- Filter by date, sender, message type (text, file, media)
- Search result highlighting with context preview
- Consider: PostgreSQL full-text search or Elasticsearch for scalability

3.5 Rich Message Formatting

- Markdown support (bold, italic, code blocks, lists)
- Syntax highlighting for code snippets
- Link previews with metadata cards (Open Graph)

3.6 Notification Controls

- Per-conversation mute and notification settings

- Do Not Disturb mode with scheduled availability
- Notification scheduling (business hours only option)

4. AI Agent Marketplace Roadmap

4.1 New AI Agents to Develop

- **Meeting Summarizer:** Records and summarizes meetings, extracts action items
- **Code Review Agent:** Reviews code changes, suggests improvements, enforces standards
- **Onboarding Agent:** Guides new employees through organization setup and policies
- **HR Assistant:** Handles leave requests, policy questions, employee documentation
- **Sales Assistant:** Tracks leads, generates reports, manages pipeline
- **Customer Support Agent:** Handles support tickets, FAQ responses, escalation

4.2 Agent-to-Agent Orchestration

- Agents that can delegate tasks to other agents
- Workflow chains (e.g., Linda creates task, Code Review agent reviews, Summarizer reports)
- Agent permissions and scoping to prevent unauthorized actions

4.3 Agent SDK/API

- Public API for third-party agent development
- Agent marketplace submission and review process
- Agent sandboxing and security isolation

5. Project Management Roadmap

5.1 Advanced Task Features

- Time tracking on tasks with reporting
- Task dependencies and blockers
- Recurring tasks with configurable schedules
- Task templates for common workflows
- Sprint planning with velocity tracking

5.2 Reporting & Analytics

- Team productivity dashboards
- Project progress reports with milestones
- Time allocation charts across projects
- Burndown charts for sprint tracking

6. Security & DevOps Risks and Recommendations -- CRITICAL SECTION

6.1 Current Security Risks

- NO end-to-end encryption -- messages are stored in plaintext in the database
- JWT tokens stored in localStorage (vulnerable to XSS attacks)
- No rate limiting on API endpoints (denial-of-service risk)
- No input sanitization audit completed
- Chat lock PINs stored in client-side storage (web: localStorage, mobile: SecureStore)
- No Content Security Policy (CSP) headers configured
- No SQL injection audit (Prisma provides some protection but not complete)
- No XSS audit on user-generated content
- WebRTC TURN credentials could be hardcoded

6.2 DevOps Improvements Needed

- CI/CD pipeline (currently manual git push + Docker rebuild)
- Automated testing (zero unit tests, zero integration tests, zero E2E tests)
- Staging environment (currently only production)
- Database backup automation
- Log aggregation and monitoring (no ELK/Grafana/etc.)
- Health check endpoints
- Container orchestration (consider Kubernetes for scaling)
- CDN for static assets
- Database connection pooling review
- Redis cluster for high availability

6.3 Code Quality Concerns

- Some TypeScript errors suppressed with 'as any' casts
- Pre-existing TypeScript errors in mobile app (missing ThemeContext module, etc.)
- No linting configuration enforced
- No code review process established
- Large component files (some > 1000 lines, should be split)
- Inconsistent error handling patterns across codebase

6.4 Compliance & Legal

- GDPR compliance not verified
- Data retention policies not implemented
- User data export (right to portability) not implemented
- Account deletion (right to erasure) not fully verified
- Privacy policy content needs legal review
- Cookie consent management not implemented

6.5 Recommended Priority Actions

1. **IMMEDIATE:** Add rate limiting and input sanitization
2. **IMMEDIATE:** Set up automated database backups
3. **WEEK 1:** Implement CI/CD pipeline with automated testing
4. **WEEK 2:** Add CSP headers and security headers
5. **WEEK 3:** JWT token rotation and secure storage
6. **MONTH 1:** Begin E2EE implementation
7. **MONTH 1:** Set up monitoring and alerting
8. **MONTH 2:** Complete security audit with penetration testing